Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Date of Assignment: 30-07-2026	Date/Time of Submission:30-07-2026

Part B - Program

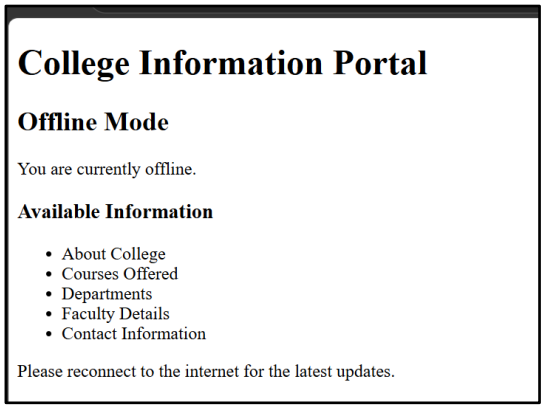
a. Develop a PWA that works in offline mode using Service Worker.

CODE:-

Manifest.json:-

```
{
 "name": "College Information Portal",
 "short_name": "College Portal",
 "description": "College website for notices, timetables and study materials.",
 "start_url": "/",
 "display": "standalone",
 "background_color": "#ffffff",
 "theme_color": "#0d6efd",
 "icons": [
  {
   "src": "icons/icon3.png",
   "sizes": "192x192",
   "type": "image/png"
  },
  {
   "src": "icons/icon4.png",
   "sizes": "512x512",
   "type": "image/png"
  }
 ]
}
```

OUTPUT:-



service-worker.js:-

```
const cache = "college-pwa-v1";
self.addEventListener("install", event => {
  event.waitUntil(
    caches.open(cache).then(cacheData => {
      return cacheData.addAll([
        "/",
        "/offline.html"
      ]);
    })
  );
  console.log("Service Worker Installed");
});
self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() => {
      return caches.match("/offline.html");
    })
```

```
    );
});
```

Offline.html:-

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Offline College Portal</title>
</head>
<body>
    <h1>College Information Portal</h1>
    <h2>Offline Mode</h2>
    <p>You are currently offline.</p>
    <h3>Available Information</h3>
    <ul>
        <li>About College</li>
        <li>Courses Offered</li>
        <li>Departments</li>
        <li>Faculty Details</li>
        <li>Contact Information</li>
    </ul>
    <p>Please reconnect to the internet for the latest updates.</p>
</body>
</html>
```

b. Create a Student Record Management PWA.

CODE:-

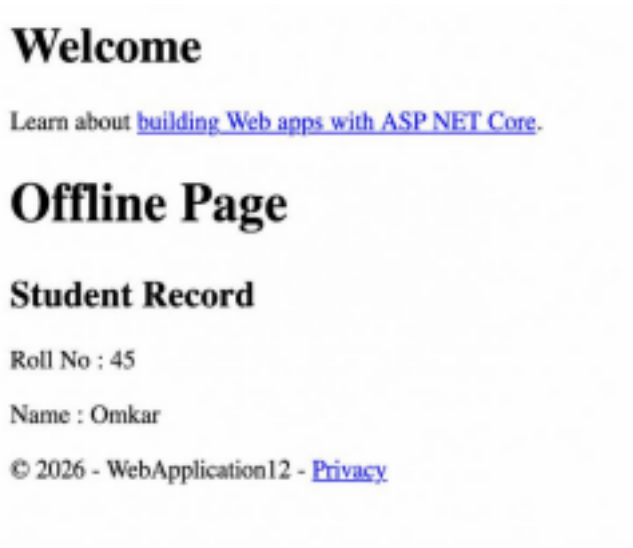
service-worker.js:-

```
const CACHE = "v1";
self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/Home/Index",
                "/css/site.css",
                "/js/site.js"
            ]);
        })
    );
});
self.addEventListener("fetch", e => {
    e.respondWith(
        caches.match(e.request).then(response => {
            return response || fetch(e.request);
        })
    );
});
```

manifest.json:-

```
{
  "name": "college portal",
  "short_name": "cp",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
```

OUTPUT:-



```
      "src": "icons/icon1.png",
      "sizes": "148x148",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

Homecontroller.cs:-

```
    public IActionResult Index()
  {
     student s = new student();
     s.RollNo = 45;
     s.Name = "omkar";
     return View(s);
  }
```

Index.cshtml:-

```
<h1>Offline Page</h1>
 <h2>Student Record</h2>
 <p>Roll No : @Model.RollNo</p>
 <p>Name : @Model.Name</p>
```

Student.cs:-

```
namespace WebApplication12.Models
{
   public class student
   {
      public int RollNo { get; set; }
      public string Name { get; set; }}}
```

